

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

What are some reasons why code might not benefit from a linear speedup when an increased number of processors are utilized?

1 / 1 point

☐

Load imbalances

☒

All of the above

☐

Costs in time due to overhead from setting up environments, calling libraries, or creating or terminating tasks

☐

Communication costs that occur between processors as they synchronize

Nice work

This is the correct answer
2.

What provides the best explanation for the difference between high performance computing and high throughput computing?

1 / 1 point

☐

High throughput computing typically involves a large number of smaller jobs while high performance computing involves a small number of larger jobs

☐

High throughput computing involves quick throughput of jobs while high performance computing provides slower throughput

☐

High throughput computing typically does not require a scheduler while high performance jobs do

☐

High throughput computing typically involves only one core while high performance computing involves a greater number of cores

Nice work

This best describes the difference between the two types of computing
3.

What is the best description of how GNU parallel works?

1 / 1 point

☐

With 10 cores available and 30 tasks, parallel will run 1 task on 1 core, with the remaining 29 tasks completing sequentially

☐

With 10 cores available and 30 tasks, parallel will run 20 tasks as first, with 2 tasks running on one core. The remaining 10 will start as the previous 20 complete

☐

With 10 cores available and 30 tasks, parallel will run 10 tasks on 10 cores, with the next set of 10 tasks running on 10 cores once the first set of 10 complete. The third set of 10 will start once the previous 10 complete.

☒

With 10 cores available and 30 tasks, parallel will run 10 tasks on 10 cores, starting each of the remaining 20 tasks as the previous 10 complete

Nice work

This is the correct answer
4.

What is the primary difference between the use of GNU parallel and Slurm array tasks?

1 / 1 point

☐

The use of Slurm array tasks is typically used with larger sized jobs, while GNU parallel with smaller jobs

☒

Using Slurm array tasks requires more input and knowledge from the user, while GNU parallel takes care of much of the parallelization automatically

☐

The use of Slurm array tasks is typically used with smaller sized jobs, while GNU parallel with larger jobs

☐

Using Slurm array tasks requires Slurm, while GNU parallel would not be used with Slurm

Nice work

This is the best answer
5.

A program takes 100 seconds to run serially on a single core. When run in parallel using 16 cores, the program completes in 10 seconds. What is the parallel efficiency of this program?

1 / 1 point

☐

16%

☐

40%

☒

62.5%

☐

100%

Nice work
6.

What is the primary benefit of using High-Throughput Computing (HTC)?

1 / 1 point

☐

Requires specialized parallel programming techniques

☒

Can run many independent computations efficiently in a short time

☐

Suitable for problems requiring frequent communication between tasks

☐

Scales linearly as the number of processors increases

Nice work

Correct. Goal of an HTC workflow is to run many independent tasks in a short amount of time, with minimal effort and overhead
7.

What is the purpose of the -J option in the GNU Parallel command?

1 / 1 point

☒

Specifies the number of tasks to run concurrently

☐

Specifies a delay between starting subsequent tasks

☐

Runs the commands in parallel across multiple nodes

☐

Creates a file containing the commands to be run

Nice work

Correct,-J option is used to specify the number of tasks to run concurrently with GNU Parallel
8.

Which of the following statements about the difference between Slurm job arrays and GNU Parallel is correct?

1 / 1 point

☒

Slurm job arrays use a separate task for each independent job, while GNU Parallel runs commands concurrently on the cores of a single node.

☐

Slurm job arrays are used for tightly-coupled parallel jobs, while GNU Parallel is used for embarrassingly parallel jobs

☐

Slurm job arrays require specialized hardware, while GNU Parallel can run on any system.

☐

Slurm job arrays are limited to running on a single node, while GNU Parallel can run across multiple nodes.

Nice work

Correct. This accurately describes the difference between the two.
9.

Which of the following statements about the efficiency of parallel computing is NOT true?

1 / 1 point

☐

Efficiency is a measure of how well the processors are doing useful work

☐

Efficiency is the ratio of the expected runtime based on the number of cores to the actual runtime

☐

If a program scales linearly, its efficiency will be 1

☒

In weak scaling, the problem size is held constant while the number of cores increases

Nice work

Correct. Weak scaling increases problem size with cores.